General-Sum Games and Correlated Equilibrium

Chen-Yu Wei

General-Sum Games

Normal-Form Game

$$[M], (\mathcal{A}_j)_{j\in[M]}, (u_j)_{j\in[M]}$$

- Player $1, 2, \dots, M$
- Player j choose action $a_j \in \mathcal{A}_j$
- Player j's payoff/utility is $u_j(a_1, \dots, a_M) = u_j(a_j, a_{-j})$

$$a_{-j} \coloneqq (a_1, a_2, \dots, a_{j-1}, a_{j+1}, \dots, a_M)$$

Two-player zero-sum game is a special case when $M = 2$ and $u_1 = -u_2$

Example: Prisoner's Dilemma

	Cooperate	Defect
Cooperate	-1,-1	-10,0
Defect	0,-10	-6,-6

Prisoner's Dilemma

	High	Low
High	5,5	0,6
Low	6,0	3,3

Pricing game

General form:

	C	D
C	x,x	z,w
D	w,z	y,y

$w > x > y > z$

Example: Chicken Game



	Chicken	Dare
Chicken	6,6	2,7
Dare	7,2	0,0

Chicken game

	Compromise	Stand firm
Compromise	0,0	-5,5
Stand firm	5,-5	-10,-10

Military confrontation / negotiation

General form:

	C	D
C	x,x	z,w
D	w,z	y,y

w > x > z > y

Example: Stag Hunt

	Stag (C)	Hare (D)
Stag (C)	6,6	2,7
Hare (D)	7,2	0,0

Stag Hunt

	Wait	Withdraw
Wait	5,5	0,4
Withdraw	4,0	2,2

Bank runs

General form:

	C	D
C	x,x	z,w
D	w,z	y,y

x > w > y > z

Equilibrium

- An equilibrium is a **steady state** where no player wants to unilaterally deviate
  - Different strategy spaces and allowed deviations give different equilibrium concepts

- Traditional game theory studies equilibria. Studying equilibria is meaningful because it allows a player to
  - Predict other players' moves
  - Reason about their own move

  It also allows the game designer (e.g. government) to
  - Design the rules of the game so that the equilibrium outcome is desirable **(mechanism design)**

Nash Equilibrium (NE)

$(P_1, P_2, \cdots, P_M) \in \Delta_{A_1} \times \Delta_{A_2} \times \cdots \Delta_{A_N}$ is called a NE if

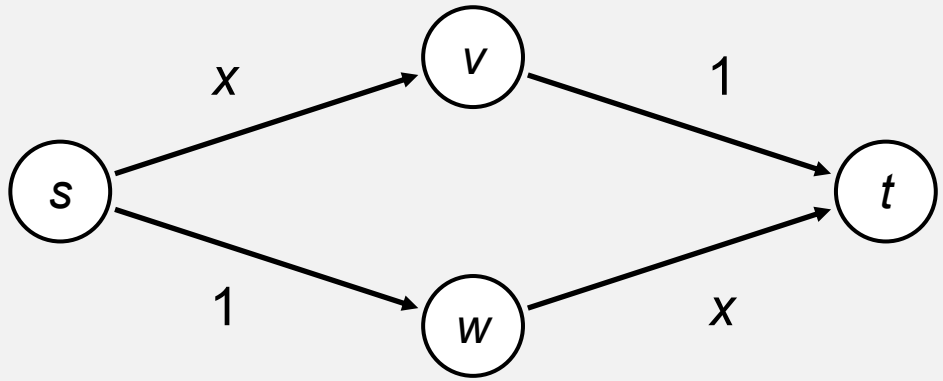
$$U_j(P_j, P_{-j}) \geq U_j(P_j', P_{-j}) \qquad \forall P_j' \in \Delta_{A_j}$$

$$U_j(P_1, P_2, \cdots, P_M) = \sum_{a_1, \cdots, a_M} P_1(a_1) P_2(a_2) \cdots P_M(a_M) U_j(a_1, \cdots, a_M)$$

NE Example: Braess's paradox

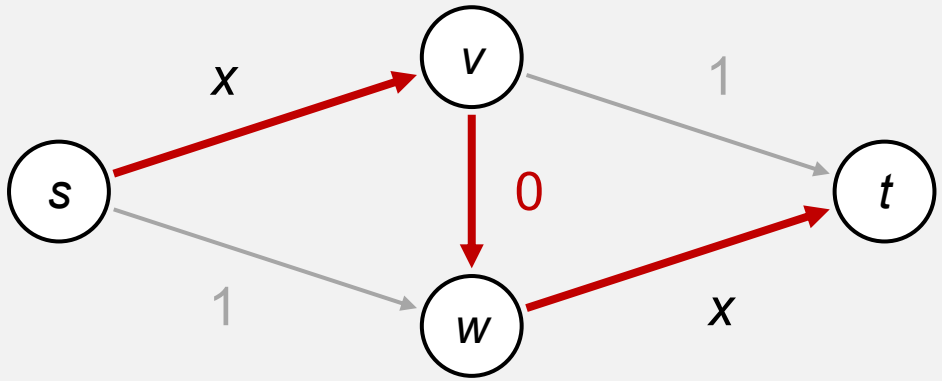
Setup: 1 unit of traffic goes from s to t, and every driver takes a fastest route.
Edge labels are travel times, where *x* = fraction of drivers using that edge.

Before



Half of the drivers take each route
Travel time = ½ + 1 = **1.5**

After: add a road v → w with travel time 0



Everyone takes s → v → w → t
Travel time = 1 + 0 + 1 = **2**

- No one wants to switch: the other two routes also take 1 + 1 = 2, so this is an equilibrium
- Everyone is worse off (1.5 → 2): each driver ignores the congestion they cause for others
- Lesson for the designer: predict the equilibrium, not just the best possible outcome

NE Examples: Second-Price Auction

Rules

- Each bidder i privately knows their value v_i
- Bidders submit sealed bids b_i simultaneously
- The highest bid wins
- The winner pays the second-highest bid

$$u_i(b_i, b_{-i}) = \begin{cases} v_i - \max_{j \neq i} b_j & \text{if } b_i > \max_{j \neq i} b_j \\ 0 & \text{otherwise} \end{cases}$$

Example (everyone bids truthfully)



Bidder 1 wins and pays 7, so their utility is 10 − 7 = 3

Why bid truthfully?

Fix bidder i (value v, bid b) and let B = $\max_{j \neq i} b_j$ be the highest competing bid.

	Truthful b = v	**Overbid b > v**	**Underbid b < v**
B < v	Win: utility v − B	Same as truthful	May lose: utility 0
B > v	Lose: utility 0	May win: utility v − B < 0	Same as truthful

In every case, b = v is at least as good, so truthful bidding is a dominant strategy

Does NE Really Reflect the Players' Behaviors

Human do not act according to NE strategy in many cases

- In the second price auction, it is found that people tends to overbid

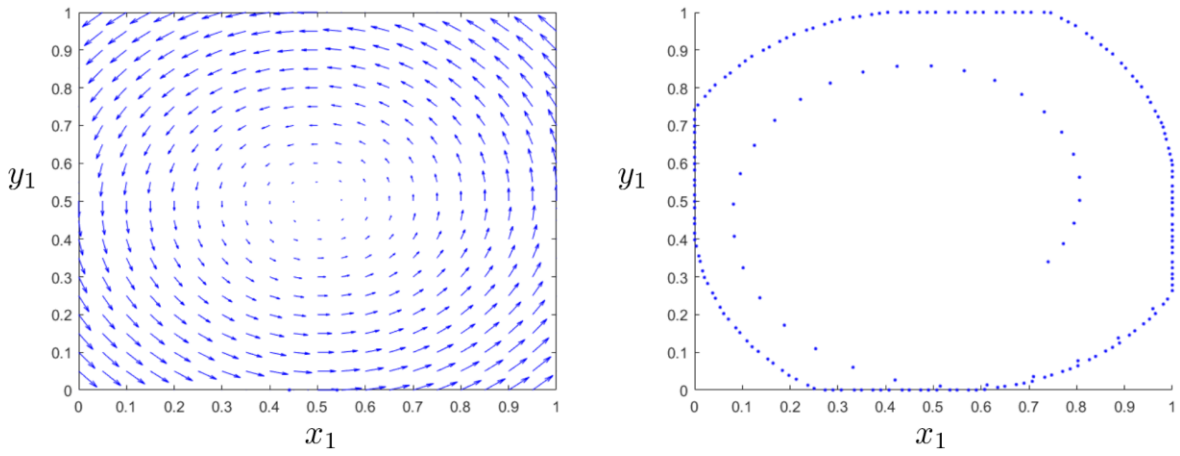
  Kagel, Harstad & Levin (1987), "Information Impact and Allocation Rules in Auctions with Affiliated Private Values: A Laboratory Study

Algorithmically, independent utility maximization algorithms do not always lead to NE

Gradient Descent/Ascent in Two-Player Zero-Sum Games

Example. Gradient descent /ascent in two-player zero-sum games leads to *divergence* from the NE.

$$x_{t+1} = x_t + \eta \nabla_x f(x_t, y_t)$$
$$y_{t+1} = y_t - \eta \nabla_y f(x_t, y_t)$$



Independent Q-Learning in Pricing Game

$$\Delta = \frac{\bar{R} - R^{\text{Bertrand}}}{R^{\text{Monopoly}} - R^{\text{Bertrand}}}$$

R^{Monopoly}: profit in full cooperation

R^{Bertrand}: profit in full competition
(i.e., equilibrium profit in one-shot game)

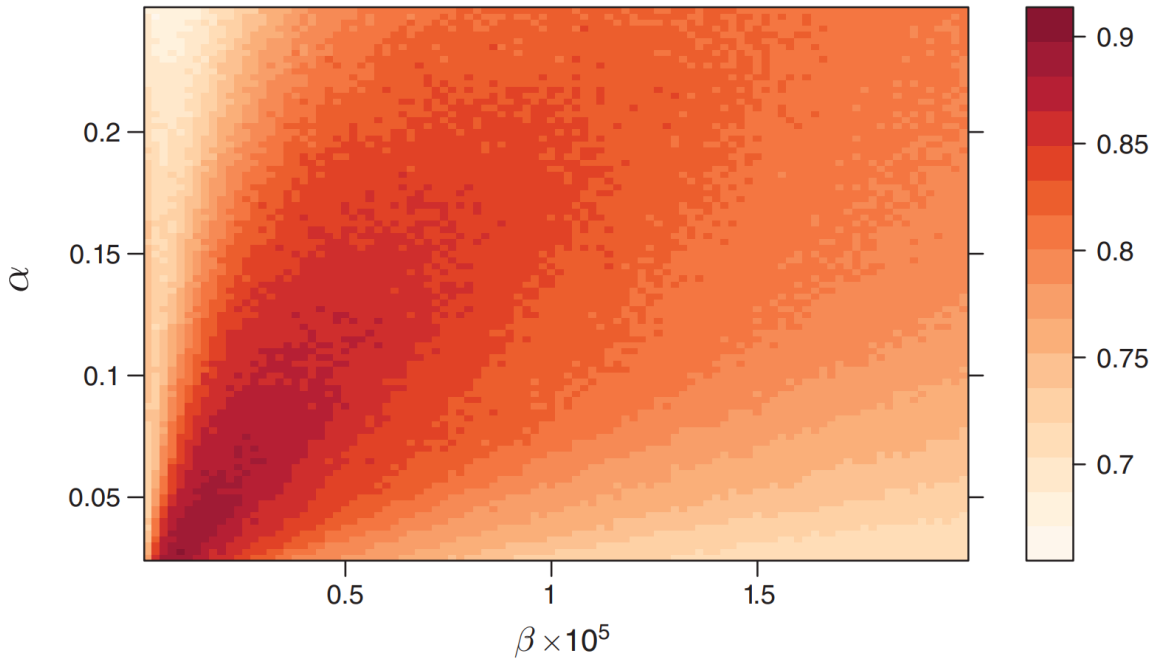


FIGURE 1. AVERAGE PROFIT GAIN Δ FOR A GRID OF VALUES OF α AND β

Correlated Equilibrium

Correlated Equilibrium: Traffic Light

Two drivers meet at an intersection (a Chicken game). A traffic light picks one of the three cells marked 1/3 at random, and shows each driver only their own light.

	Stop	**Go**
Stop	6, 6 prob. 1/3	2, 7 prob. 1/3
Go	7, 2 prob. 1/3	0, 0 prob. 0

Payoffs are (driver 1, driver 2); driver 1 picks the row. Red: probability that the light picks the cell.

	Payoff per driver	**Crash prob.**
Mixed NE	14/3 ≈ 4.67	1/9
Traffic light	**(6 + 7 + 2)/3 = 5**	**0**

The light shows you green (Go)

The other driver must have red, so they stop
Obey (Go): 7 Deviate (Stop): 6 → **obey**

The light shows you red (Stop)

The other driver has red or green, 1/2 each
Obey (Stop): ½·6 + ½·2 = 4
Deviate (Go): ½·7 + ½·0 = 3.5 → **obey**

No driver wants to disobey their own light, so this distribution σ is a **correlated equilibrium**:

$$\mathrm{E}_{a\sim\sigma}[\, u_i(a) \mid a_i \,] \;\geq\; \mathrm{E}_{a\sim\sigma}[\, u_i(a_i', a_{-i}) \mid a_i \,] \quad \text{for all } i, a_i, a_i'$$